Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/14/2005
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Project Title:	Experimental Investigation of Impurity Effects of the Two-Phase Isochoric Heat Capacity Near the 3He Critical Point		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Barmatz, Martin (JPL) Zhong, Fang (JPL)		
Grant/Contract No.:	NAG3-2901		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Task Description:	The famous Yang-Yang thermodynamic relation establishes a direct connection between the singularity in the heat capacity at constant volume in the vicinity of a liquid-gas critical point and the temperature dependence of the second derivatives in the chemical potential and the pressure. An unresolved question that has never been satisfactorily answered is: to what extent is the specific heat singularity “shared” between the pressure and the chemical potential? All previous theories and experiments have suggested that there is no singular contribution from the chemical potential variation. Moreover, all currently available equation-of-state models are based on this assumption. However, this assumption was recently challenged by prominent theorists who argued that the pressure and the chemical potential, in fact, contribute almost equally to the specific heat singularity (“Yang-Yang anomaly”). In this project high-precision isochoric heat capacity measurements will be performed along many near-critical isochores in the two-phase region of pure 3He and in the same system with a perfectly controlled impurity level in order to unambiguously determine the behavior of the second derivative of the chemical potential with respect to temperature. The unique property of 3He is that the only important impurity in the sample is 4He, which can easily be reduced to 1 ppm. Moreover, 3He is the most symmetrical with respect to the critical isochore of all fluid systems and the most quantum-mechanical of all fluids (i.e., it has the largest DeBoer parameter). This means the asymmetry effect will not mask the Yang-Yang anomaly (if it exists) and the large DeBoer parameter could possibly make this anomaly more pronounced. Such measurements will be repeated for different additives of 4He into the sample. The thermodynamic analysis of the impurity effects will be used to compare the experimental results with the theoretical predictions. Our experiments should provide precursor evidence of any divergences in the second derivatives of the pressure and chemical potential. These measurements will also produce reliable thermodynamic information on the near-critical properties of 3He and 3He-4He mixtures. Gravity effects will distort ground-based measurements very near the critical point. We plan to use the latest equation-of-state models to correct data close to the transition that experience a few percent corrections due to gravity. This investigation, that will take advantage of the high precision sensors developed for the MISTE flight experiment, should be able to perform measurements an order of magnitude closer to the transition than in previous 3He studies. The objective of this project is to resolve one of the most fundamental questions of the critical state of fluids, the so-called “Yang-Yang dilemma”, and produce reliable thermodynamic information on near-critical properties of 3He and 3He-4He mixtures. One of the NASA program rationales for the Fundamental Physics discipline is to utilize the unique environment of space “where extremely uniform samples free from hydrostatic compression are required.” A microgravity experiment will be necessary to perform gravity-free measurements sufficiently close to Tc to unambiguously resolve the Yang-Yang dilemma. This future flight experiment could be flown in the Low Temperature Microgravity Physics Facility (LTMPF) using the same experimental cell being developed for the MISTE flight experiment. If confirmed, the Yang-Yang anomaly would require essential revision of the scaling theory for near-critical fluids and reformulation of equations-of-state models that are currently being used in engineering applications. In particular, it will have a crucial impact on understanding of the legitimacy of the lattice-gas model for real fluids.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our analysis of earlier thermodynamic measurements near the 3He liquid-gas critical point in light of solving the Yang-Yang dilemma has been published in the J. Low Temp. Phys. That study revealed subtle gravity effects (previously ignored) that require the careful placement of the pressure sensor within the experimental cell. The existing pressure sensor has been repositioned within the experimental cell to minimize gravity effects. New highly stable PdMn high-resolution thermometers and a density sensor similar in design to the proposed MISTE flight sensor have also been fabricated. The MISTE experimental cell and low temperature probe are now being modified in preparation for measurements of the specific heat along 16 isochores both above and below the critical point. Drift as well as pulsed specific heat measurements will be performed. The drift measurements will determine the shape of the coexistence curve using the quasi-static thermogram method that was recently applied to 3He (I. Hahn, M. Weilert, F. Zhong, and M. Barmatz, High-Resolution Measurements of the Coexistence Curve Near the 3He Liquid-Gas Critical Point,” J. Low Temp. Phys. 137, 579, (2004)). A new probe is also being fabricated to provide better thermal isolation. This new probe will be used in the future to perform high-resolution pressure measurements along the critical isochore to further test the Yang-Yang expression close to the critical point. An advanced data analysis approach has also been developed to analyse our specific heat measurements. That approach was recently used in the reanalysis of SF6 microgravity specific heat measurements (M. Barmatz, F. Zhong , and A. Shih, “Reanalysis of Microgravity Heat Capacity Measurements Near the SF6 Liquid-Gas Critical Point,” Int. J. of Thermophys. 25, 1667, (2004)).
Bibliography Type:	Description: (Last Updated: 08/06/2008)
Articles in Peer-reviewed Journals	Anisimov, M.A.; Zhong, F.; and Barmatz, M. "Resolving the Yang-Yang Dilemma in 3He Near the Critical Point" J. Low Temp. Phys. 137, 69 (2004) , Oct-2004